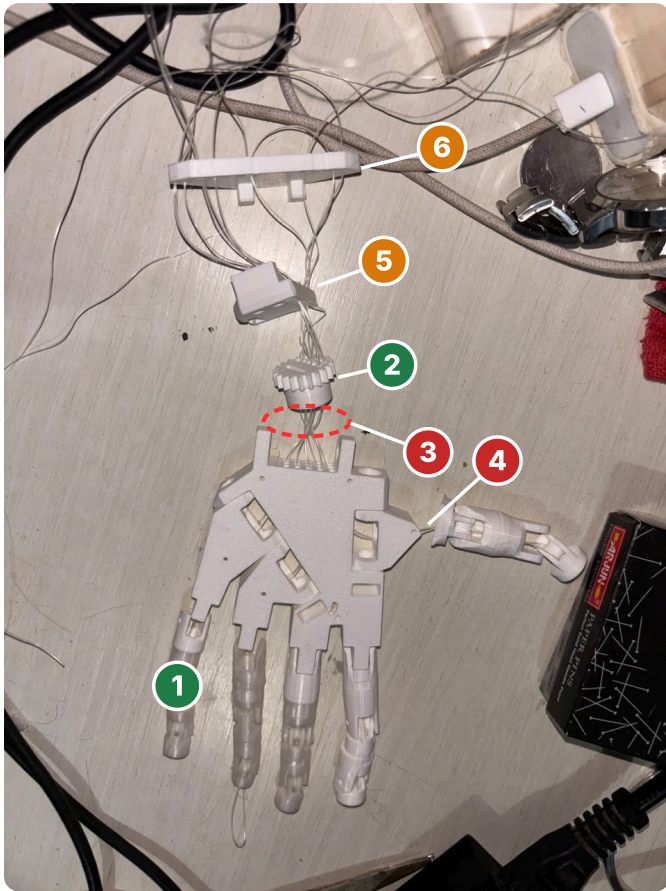


Somatic Arm: finish the assembly

Starts from your photo (7 Oct). 10 steps. **Red** = closing string (palm side). **Blue** = opening string (back). Hand back up, thumb on your left.



Your photo, marked up.

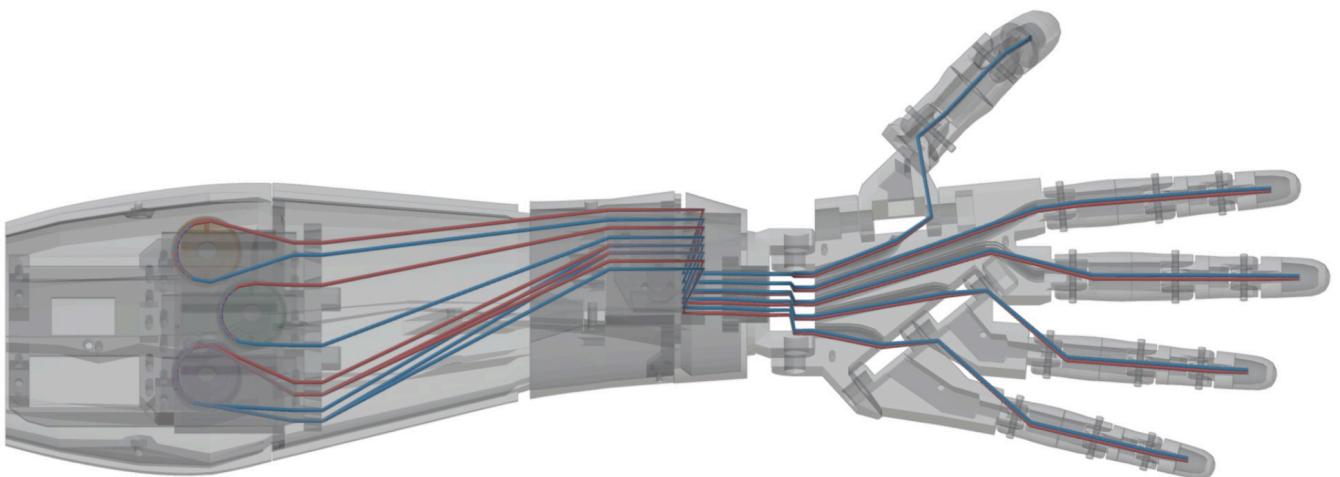
- 1 Fingers strung.** 10 ends out of the palm grid. Good.
- 2 Gear on,** round hub toward the hand. Good.
- 3 Ring 3 is missing here.** It must sit between the palm and the gear. Fix in **Step 1**.
- 4 Thumb is loose,** hanging on its line. Fix in **Step 2**.
- 5 6 Block and flat piece:** slide them off the strings. Only ring 3, gear, ring 2 go on now (**Step 1**).

You need:

- Fevikwik + toothpick
- Bamboo skewer
- Masking tape
- Marker pen
- Scissors, tweezers
- Laptop + Uno + USB
- Servo power (AA pack)
- Your assembly guide

The 10 steps

- | | | | | |
|---------------------------------------|---|-------------------------------------|--------------------------------|-------------------------|
| 1 Fix the order on the strings | 2 Fix the thumb to the palm | 3 Check the forearm is ready | 4 Ring 1, then the comb | 5 String guide |
| 6 Strings to the servos | 7 Slide wrist home, glue, skewer | 8 Servos to 0°, pulleys on | 9 Tie the strings on | 10 Test and tune |

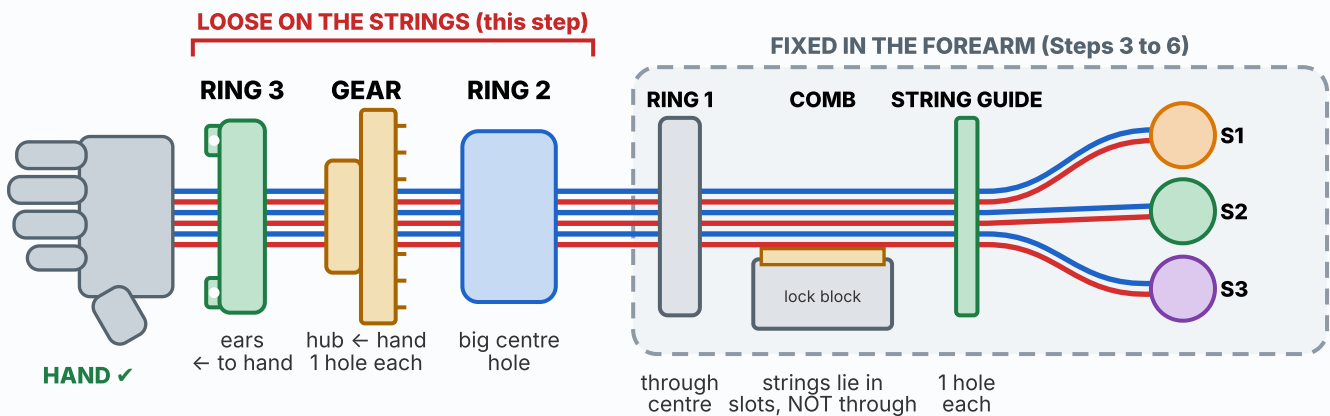


The goal: the finished arm from above. Servos and pulleys at the elbow end (left), hand on the right.

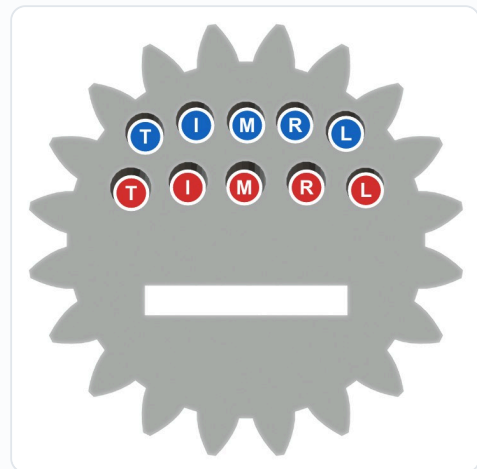
1

Fix the order on the strings

NO GLUE YET



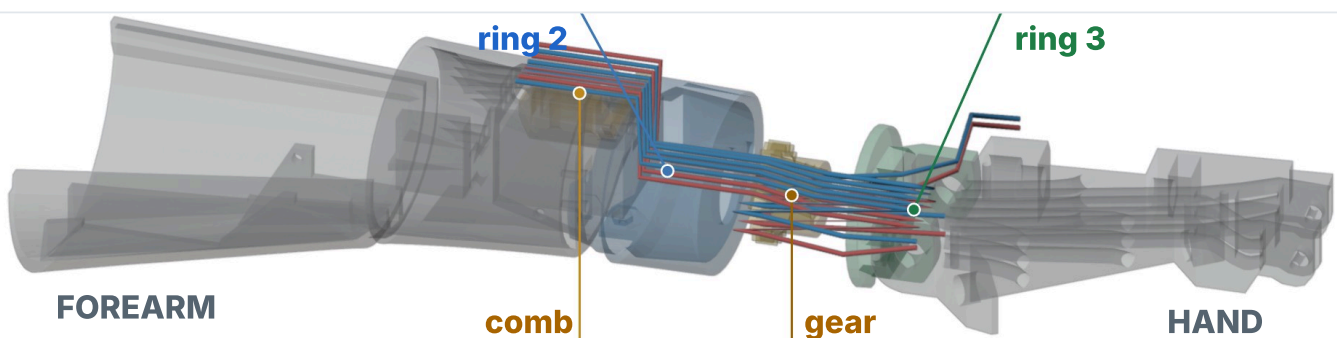
1. **Slide off** everything on the **loose ends**: the block, the flat piece, the gear. The hand stays strung.
2. Thread **ring 3**: all 10 ends through its big centre hole. **Ears toward the hand**.
3. Thread the **gear**: each end through **its own hole** (map). **Round hub toward the hand**, teeth toward the forearm.
4. Thread **ring 2**: all 10 through its big centre hole.
5. Leave all three loose. **No glue**.



Gear seen from the **teeth** side. **Top row blue, bottom row red**. T I M R L, left to right.

Always slide parts on and off from the loose ends. Never pull strings back out of the palm.

✓ **DONE WHEN:** from the palm: ring 3 → gear → ring 2 → loose ends. Nothing else on the strings.



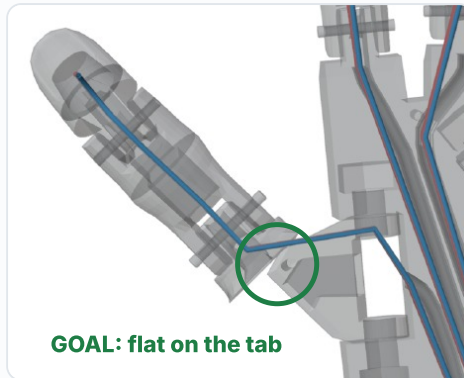
Same order in 3D, side view (from your stringing guide). Hand on the right here.

2 Fix the thumb to the palm

FEVIKWIK DOTS



Your photo: the thumb's wide end is off the palm.



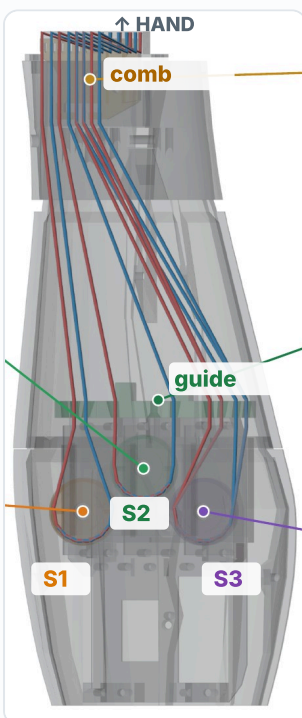
Goal: the wide end sits flat on the angled tab of the palm.

1. Push the thumb's **wide (funnel) end** flat against the **angled tab** on the palm. Its line runs straight from the tab into the funnel.
2. Put **2 or 3 tiny Fevikwik dots** on the **outer edge** of the joint, with a toothpick. **Nowhere near the centre hole** where the line runs.
3. Hold 30 seconds. Point the thumb the same way as in the goal picture.

✓ **DONE WHEN:** thumb is solid on the palm, and pulling its **red** end still bends it.

3 Check the forearm is ready

CHECK



Forearm from above (lid off). Hand end at the top.

- ☐ **3 servos in place.** Hand end away from you:
S1 thumb left, **S2 index** middle, **S3 middle + ring + little** right.
- ☐ **String guide** (green) in place, between the comb and the servos.
- ☐ **Ring 1** glued on the wrist end of the forearm.
- ☐ **Lock block + comb** glued at the wrist end (comb on top of the block).
- ☐ **Pulleys OFF** for now. They go on in Step 8.

Any box empty? Do it now with your assembly guide, steps up to 21. Then come back to Step 4.

Label each servo cable with tape: **S1, S2, S3**. You need this in Step 8.

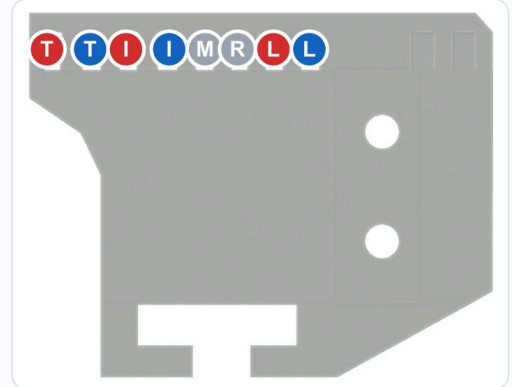
4 Through ring 1, then into the comb

COMB

1. Hold the forearm with the **lid side up**.
2. Push all **10 loose ends** through **ring 1's centre hole**, into the forearm.
3. Go **over** the lock block, not through it.
4. Press each string down into **its slot** in the comb. Left to right:

T T I I M R L L

Grey = that finger's red and blue share one slot (middle, ring).

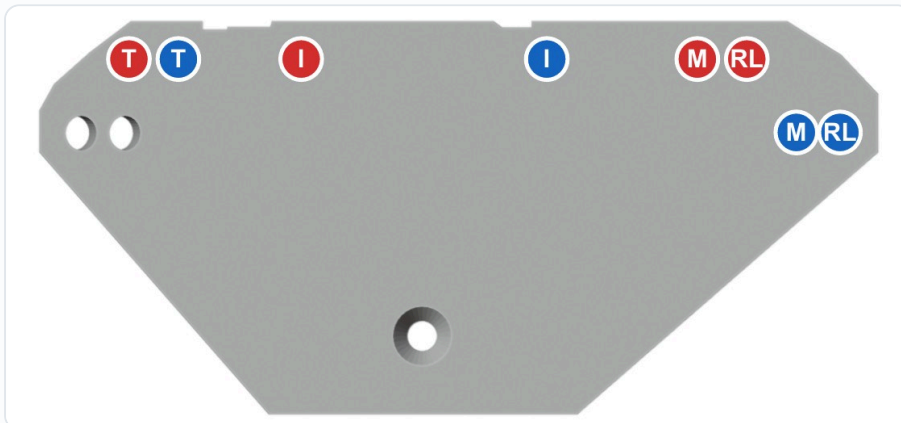


Comb from above. Hand end at the top. Thumb on the left.

✓ **DONE WHEN:** 10 strings through ring 1, each lying in its comb slot, none crossing.

5 Through the string guide

STRING GUIDE



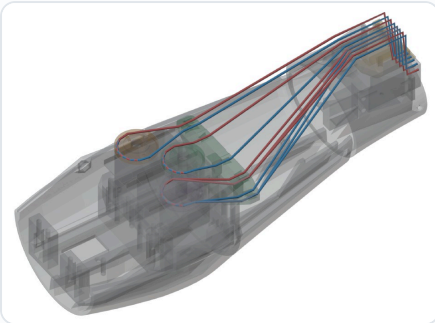
String guide, looking toward the hand. Thumb on the left. Unmarked holes are spare.

1. Push each string through **its own hole** in the green string guide, toward the servos.
2. **Thumb** holes on the left. **Index** in the middle. **Middle + ring + little** on the right.
3. **Ring and little share a hole:** one for both reds **RL**, one for both blues **RL**.
4. Pull each string gently. If it rubs or crosses another, swap its hole.

✓ **DONE WHEN:** every string goes straight from its comb slot to its guide hole. No crossings.

6 Bring each pair to its servo

DON'T TIE YET



Strings from the wrist (right) to the 3 pulleys.

STRINGS

GO TO

T **T** thumb (2)

S1, left

I **I** index (2)

S2, middle

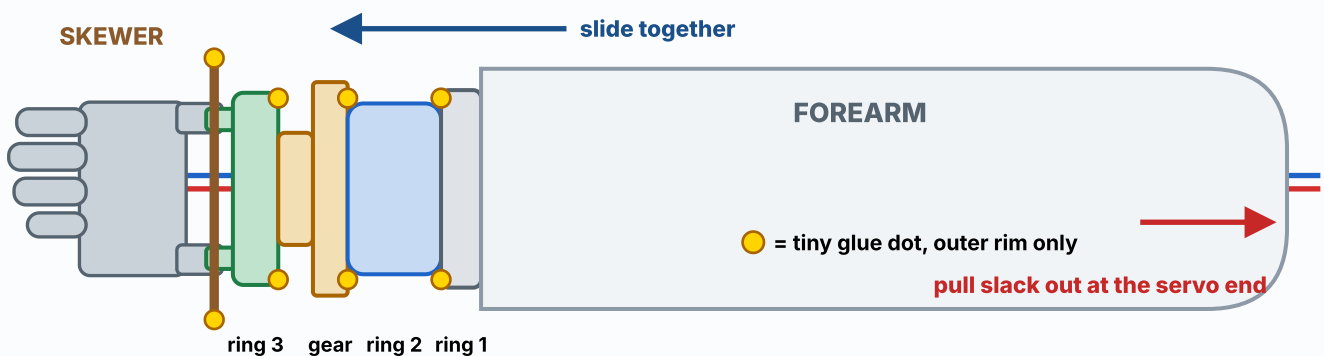
M **M** **R** **R** **L** **L** (6)

S3, right

1. Let each group hang next to its servo.
2. Tape each group to the forearm wall with masking tape, so they don't mix.
3. **Don't tie.** Keep at least 15 cm spare on each.

7 Slide the wrist home, glue, fix the hand

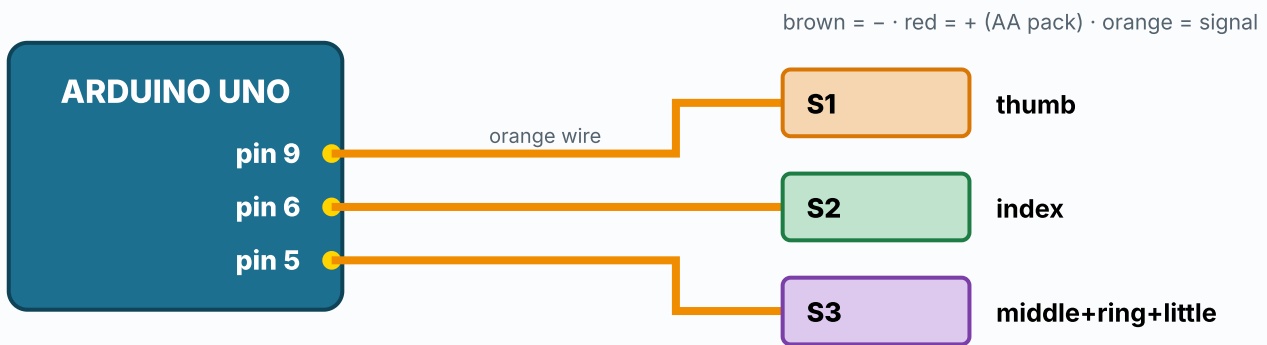
GLUE: DOTS ONLY



1. Slide **ring 2, gear, ring 3** up against ring 1. At the same time **pull the loose ends** at the servo end, so no loops get trapped.
2. Hand fully open. Put **tiny Fevikiwik dots** on the **outer rim** where each part touches the next. Keep glue away from the centre and the strings.
3. Slide the hand up to **ring 3**. Line up the holes in ring 3's ears and the palm's wrist tabs (as in assembly guide step 23).
4. Push the **skewer** through all the holes. Cut it 2 mm past each side. One dot of glue on each skewer end only.

Never let glue touch a string. A glued string can't slide, and that finger is dead.

✓ **DONE WHEN:** pull each of the 10 strings at the servo end: every one still slides and moves its finger.



1. Plug the servo cables: **S1 → pin 9, S2 → pin 6, S3 → pin 5**. Power as in your wiring guide (AA pack on the servos, grounds joined).
2. In Arduino IDE open `firmware/servo_zero/servo_zero.ino` and **Upload**. Electrodes OFF.
3. All 3 servos go to **0°** and stay there. (They twitch once after 3 s. Ignore it for now.)
4. Push each pulley **straight down** onto its servo. **Don't turn the shaft**. Turn the pulley so its **tie hole points to the elbow** (away from the hand).
5. Press the Uno's **RESET** button. After 3 s, S1, then S2, then S3 turns a little toward **close** and back.
6. Watch each one. **Draw an arrow** on top of each pulley in the way it turned.



Pulley sits on a servo horn? Glue the horn into the pulley first (Fevikwik, wait 2 min). Then push it on.

Missed the turn? Press RESET again. It repeats every time.

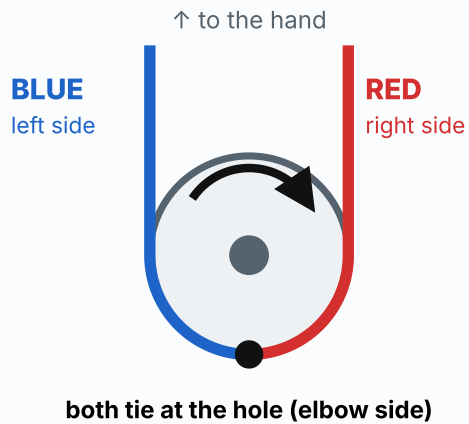
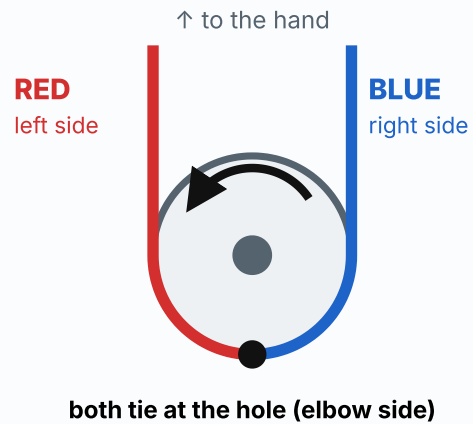
✓ **DONE WHEN:** 3 pulleys on, servos at 0°, an arrow on each pulley showing "close".

9

Tie the strings to the pulleys

HAND OPEN

Look down at the pulley, **hand end at the top**. Pick the picture that matches **your arrow**:

Arrow turns **CLOCKWISE**Arrow turns **ANTICLOCKWISE**

Why: the "close" turn winds **red** onto the pulley (finger bends) and lets **blue** off at the same time.

1. **Check first:** pull the **red** end by hand, the finger closes. Pull **blue**, it opens. If not, they're swapped: swap the colour tape.
2. All fingers **straight**. Servo still at 0° (servo_zero running).
3. **Red:** down its side, around the pulley groove, through the tie hole (no hole: round the horn arm). Pull until the slack is gone but the finger **does not bend**. Tie 3 knots.
4. **Blue:** down the other side, through the same hole. Pull **just snug**. Tie 3 knots.
5. One tiny drop of Fevikwik on each **knot** only. Cut the ends to 1 cm.

S3 has 6 strings. Tie the 3 reds (M, R, L) one at a time, each just snug, then the 3 blues. All 3 fingers must start bending together.

✓ **DONE WHEN:** S1, S2, S3 tied. Hand stays fully open. No string is slack.

10 Test and tune

TEST

A. Servo test (no muscle sensor needed)

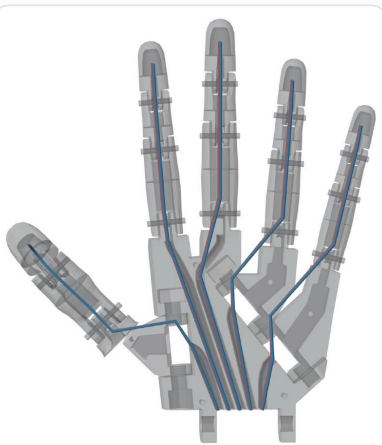
1. Upload `firmware/servo_test/servo_test.ino`.
Electrodes OFF.
2. It closes and opens **S1**, then **S2**, then **S3**, one at a time.
3. Then all 3 close and open together, 5 times. It repeats.

B. Real use (muscle sensor)

1. Upload `firmware/somatic_arm/somatic_arm.ino`.
Unplug USB.
2. Electrodes on. LED **solid**: relax 3 s. LED **blinking**: squeeze hard 3 s. LED **off**: ready.
3. Squeeze: hand closes. Relax: it opens.

IF YOU SEE THIS	DO THIS
Finger opens when its servo says close	Red and blue are swapped on that pulley. Re-tie them on the opposite sides.
Finger closes only a little	Re-tie its red 2 to 3 mm tighter. Still short: raise that servo's <code>CLOSED_ANGLE</code> by 5.
Servo buzzes or gets hot when closed	Lower that servo's <code>CLOSED_ANGLE</code> by 5 (e.g. 90 → 85). Repeat until quiet.
Finger stays bent when open	Blue too loose. Re-tie it snug.
One of M / R / L lags the others	Its red is looser. Re-tie it a little tighter.
String slips on the pulley	Re-knot through the hole, 3 knots, drop of Fevikwik on the knot.
Uno restarts mid-test (Serial Monitor shows "servo test start" again)	Weak AA cells or a loose power wire. Fresh cells, check the breadboard wires.

Where to change angles: line `const int CLOSED_ANGLE[3] = {90, 90, 90};` Order is S1 thumb, S2 index, S3 M+R+L. Change it the same way in **somatic_arm.ino**, **servo_test.ino** and **servo_zero.ino**.



✓ **DONE WHEN:** all 5 fingers close and open with the servos, 10 times in a row, no buzzing.

Last: close the forearm lid. Use tape, not glue, so you can re-tune before the evaluation.

Upload code only with the electrodes OFF your arm (or the laptop charger unplugged).